



By Nitesh Singh

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SOC

Analyst L1 For Free

Defronix Cyber Security → <https://defronix.com>

Router



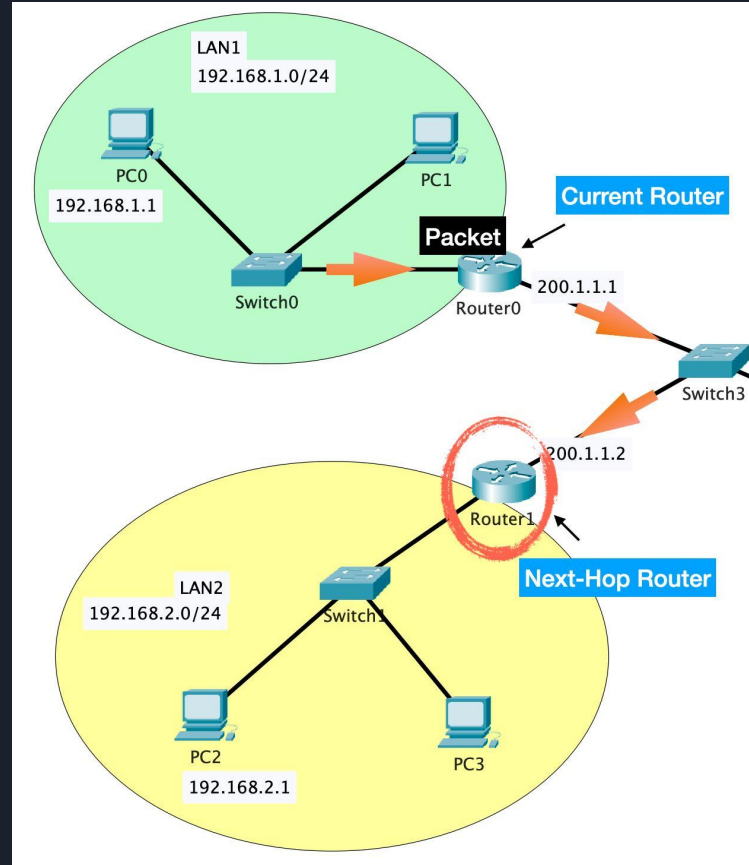
What is a Router ?

A router connects **different networks**.

LAN → Router → Internet

What Router Does ?

- Routes packets based on IP
- Performs NAT (Private → Public IP)
- Maintains routing table.





Router

SOC Logs from Router

- Source IP
- Destination IP
- Traffic flow

SOC Indicators

- Traffic to malicious IP
- Unusual outbound connections
- Data exfiltration

Switch



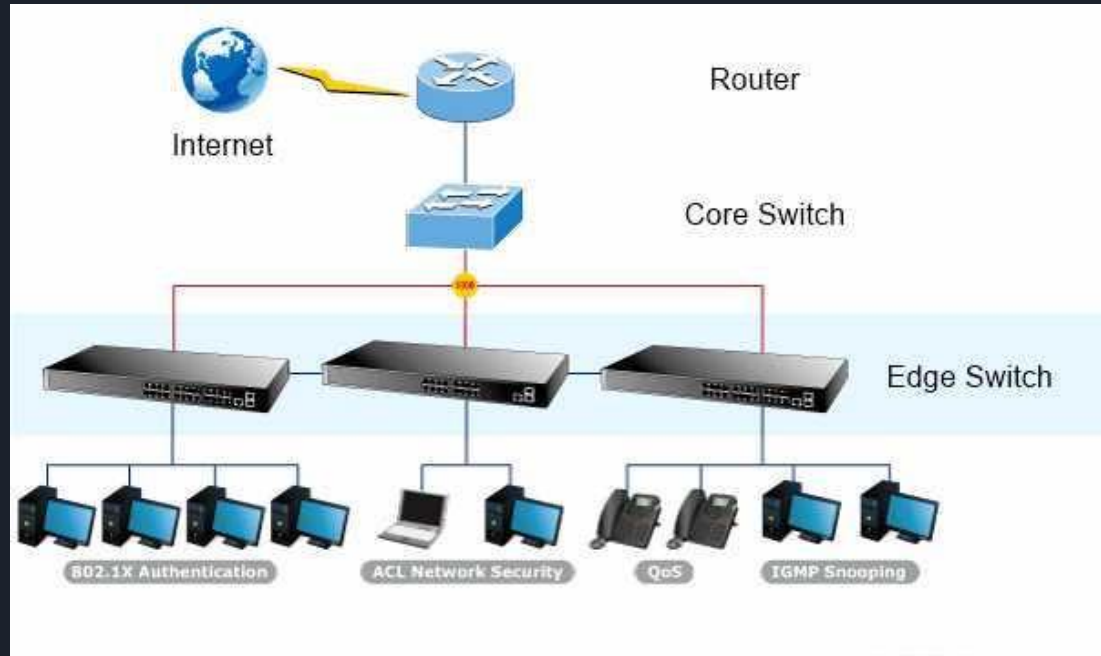
What is a Switch?

Connects devices **inside** a network (LAN).

PC ↔ Switch ↔ Server

What Switch Does ?

- Uses MAC addresses
- Sends data only to correct device





SOC Perspective

- Detect internal movement
- MAC anomalies

Threats

- ARP spoofing
- MAC flooding

SOC Example

Same MAC seen on multiple ports

Possible:
MITM / ARP spoofing



VPN (Virtual Private Network)

What is a VPN?

A VPN creates a **secure encrypted tunnel** between a user and a network.

User → Encrypted Tunnel → VPN Server → Internet / Internal Network

What VPN Does ?

- Encrypts traffic (Confidentiality)
- Hides real IP (Anonymity)
- Allows remote access (Corporate use)



How VPN tunneling works

→  Encrypted →  Unencrypted



Device

VPN server

Websites
or apps

VPN (Virtual Private Network)



Types of VPN

Remote Access VPN

- Employee → Company network
- Example: Work from home

Site-to-Site VPN

- Office A ↔ Office B
- Used in enterprises

SSL VPN

- Uses HTTPS (port 443)
- Common in corporate setups

SOC Perspective

Normal Behavior

User IP → VPN IP → Internal systems



Suspicious Indicators

- Login from unusual country
- Multiple logins (same user, different locations)
- VPN used + suspicious activity after login
- High data transfer through VPN

Common VPN Abuse

Attackers use VPN to:

- Hide identity
- Bypass geo restrictions
- Access internal network

SOC Example

User: admin

Login IP: Germany

VPN IP: India

Time gap: 2 minutes

Impossible travel → **Compromised account**

Firewall



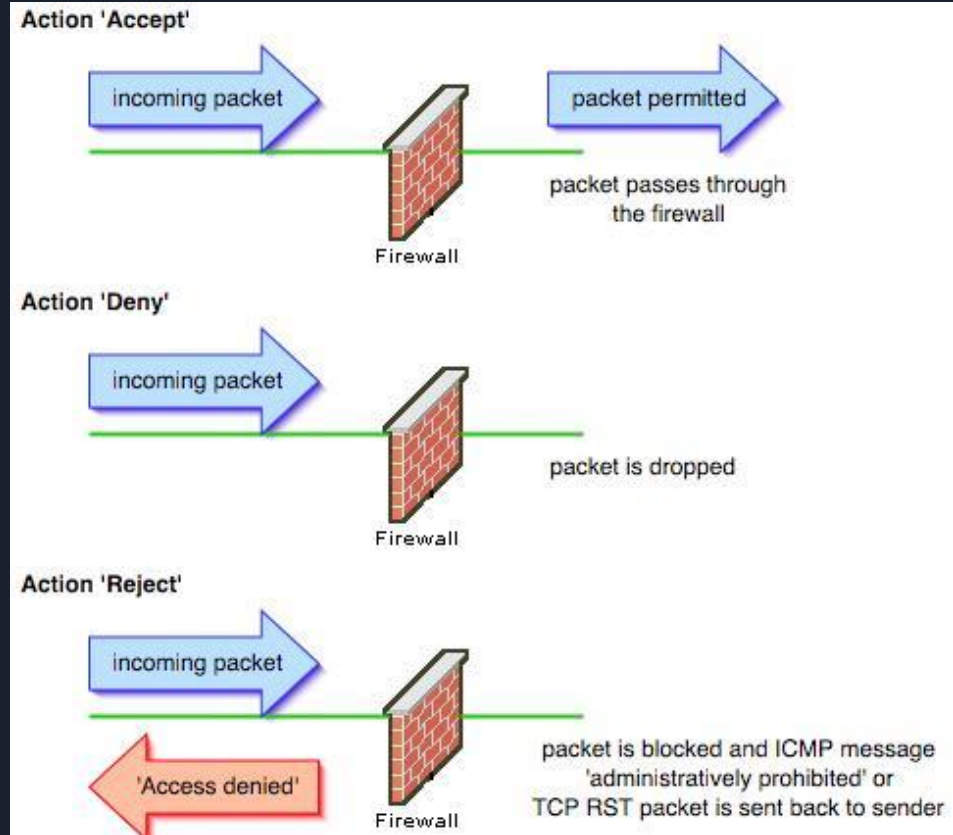
What is Firewall?

Controls traffic based on rules.

Allow / Deny traffic

Types

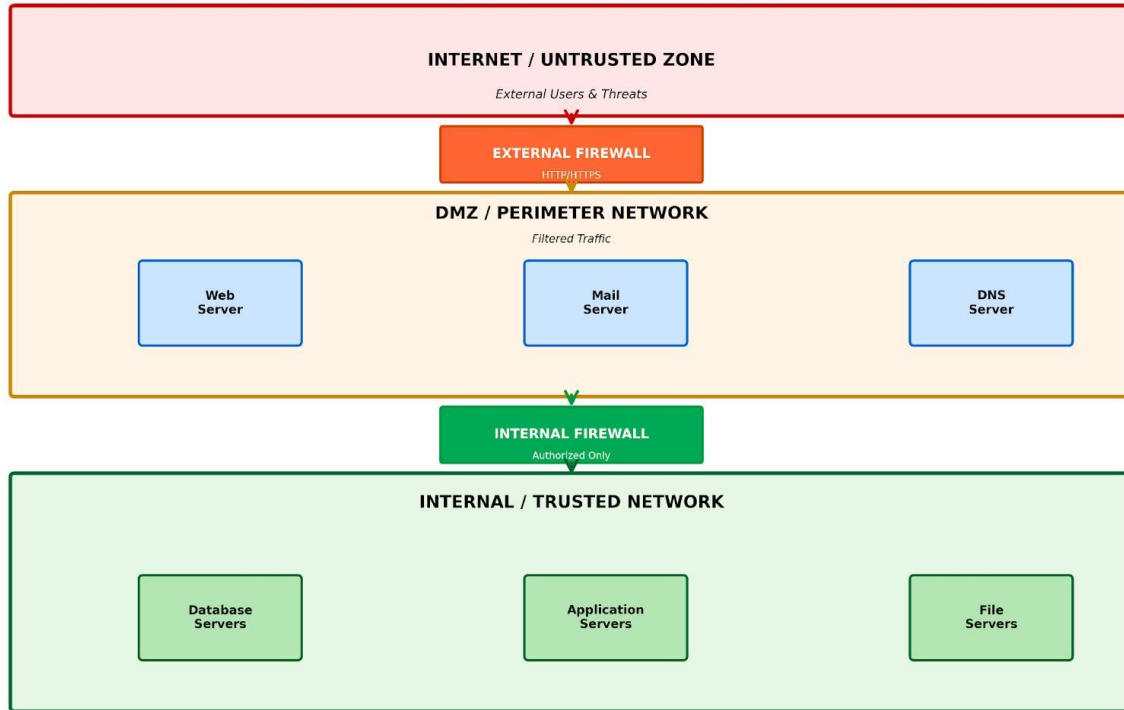
- Network Firewall
- Host Firewall
- Stateful Firewall





Firewall

DMZ Network Segmentation



✓ External firewall blocks unauthorized inbound traffic

✓ Internal firewall prevents DMZ compromise from spreading to internal network

SOC Indicators

- Blocked attack attempts
- Port scanning
- Suspicious outbound traffic

Example

Port 3389 attempts: 500
Action: Denied

RDP brute force attempt



IDS (Intrusion Detection System) and IPS (Intrusion Prevention System) are network security tools that protect against cyber threats.

IDS passively monitors traffic and alerts on suspicious activity, while IPS sits inline to actively drop malicious packets and block threats in real-time.

They are often integrated into **Next-Generation Firewalls** or **Unified Threat Management** (UTM) devices.

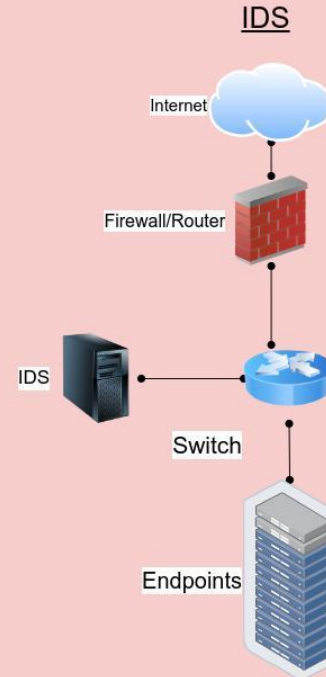
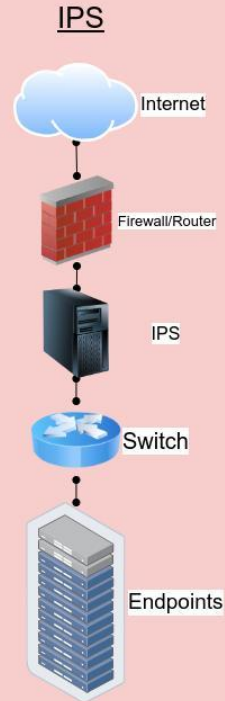
IDS (Intrusion Detection System)

- Detects attacks
- Alerts SOC

IPS (Intrusion Prevention System)

- Detects + blocks attacks

IDS / IPS





SOC Alerts

- SQL Injection
- Malware traffic
- Exploit attempts

SOC Challenge

- False positives
- Alert fatigue



Proxy Servers

Acts as **middleman** between user & internet

User → Proxy → Internet

Proxy vs VPN ??

The core difference is that VPNs encrypt all internet traffic at the operating system level, offering high security and privacy, while proxies typically do not encrypt data and only work at the application level (e.g., browser-only).

VPNs are ideal for total security, whereas proxies are faster and better for simple IP masking or bypassing geo-restrictions.

VPNs provide robust encryption (e.g., AES-256), making traffic unreadable to ISPs and hackers. Proxies generally provide no encryption, meaning data is less secure.



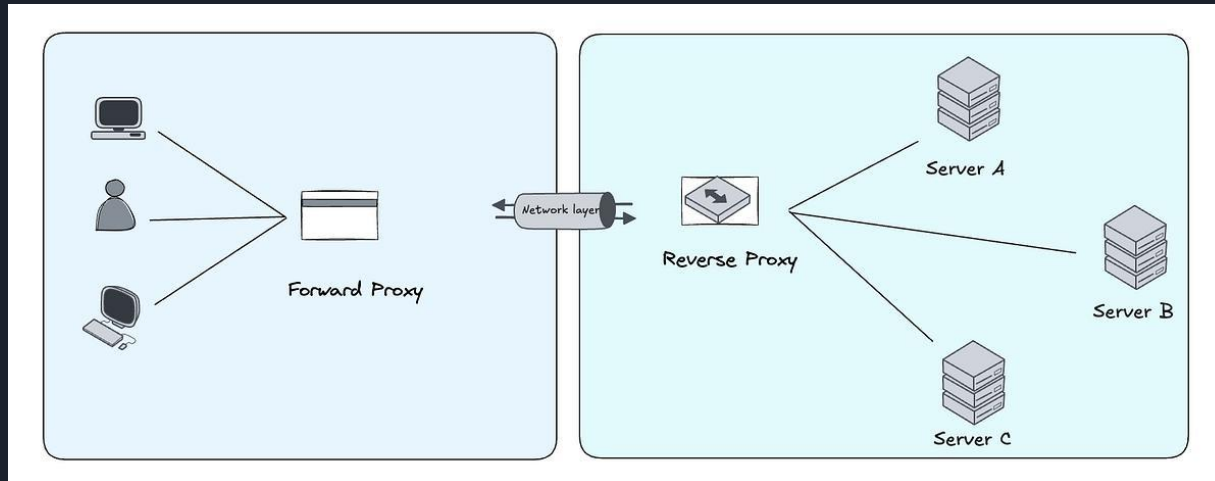
Proxy Servers

Forward & Reverse Proxies

Forward proxies sit in front of clients to manage outbound requests, providing anonymity and content filtering,

while **Reverse proxies** sit in front of servers to handle incoming requests, managing load balancing, caching, and security.

Forward proxies serve clients (users), whereas reverse proxies serve origin servers.





SOC Logs

- URLs accessed
- User activity
- Download/upload

SOC Indicators

- Access to malicious domains
- Data exfiltration
- Unusual browsing



Proxy Servers

Flow after combining all together now

User → Switch → Router → Firewall → Proxy → Internet

Device	What SOC Sees
Router	IP traffic
Switch	MAC activity
Firewall	Allowed/blocked traffic
IDS/IPS	Attack alerts
Proxy	User behavior

Proxy Servers



Real SOC Scenario

User downloads file from unknown domain

Firewall: Allowed

Proxy: Suspicious URL

IDS: Malware detected

Conclusion:

Malware infection attempt

Analyst Thinking

When alert comes, ask:

Where did this happen?

- Internal → Switch / MAC
- Network → Router / IP
- Security → Firewall / IDS
- User → Proxy